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CLASS	BSCS
SEMESTER	1st
SECTION	(A)
SUBJECT	C++
SUBMITTED TO	SIR ASAD
ASSIGNMENT LAB	3

Task_01

```
#include <iostream>
 using namespace std;
 int main () {
    int age;
    double pay;
    char section;
cout << "Enter age, pay and section:";
 cin >> age >> pay >> section;
 cout << "Age: " << age << endl;
 cout << "Pay: " << pay << endl;
 cout << "section: " << section << endl;

 return 0;
}
```

output:-

```
Enter age, pay and section: 20 2500.5A
Age: 20
pay: 2500.5
Section: A
```

```
#include <iostream>
using namespace std;
int main (){
    int age;
    double pay;
    char section;
    cout<<"Enter age,pay and section:";
    cin>>age>>pay>>section;
    cout<<"Age:"<<age<<endl;
    cout<<"Pay:"<<pay<<endl;
    cout<<"Section:"<<section<<endl;
    return 0;
}
```

```
Enter age,pay and section:20 2500.5 A
Age:20
Pay:2500.5
Section:A

[Program finished]
```

Task-02

```
#include<iostream>
  using namespace std;
   int main()
{
   cout<<"28/4-2="<<28/4-2<<endl;
   cout<<"6+12*2-8="<<6+12*2-8<<endl;
   cout<<"4+8*2="<<4+8*2<<endl;
   cout<<"6+17%3-2="<<6+17%3-2<<endl;
   cout<<"2+22*(9-7)="<<2+22*(9-7)<<endl;
   cout<<"(8+7)*2="<<(8+7)*2<<endl;
   cout<<"(16+7)%2-1="<<(16+7)%2-1<<endl;
   cout<<"12/(10-6)="<<12/(10-6)<<endl;
   cout<<"(19-3)*(2+2)/4="<<(19-3)*
   (2+2)/4<<endl;
    return 0;
  }
```

Output :-

```
28/4-2=5
6+12*2-8=22
4+8*2=20
6+17%3-2=6
2+22*(9-7)=46
(8+7)*2=30
(16+7)%2-1=0
12/(10-6)=3
(19-3)*(2+2)/4=16
```

```
#include <iostream>
using namespace std;
int main (){
    cout<<"28/4-2="<<28/4-2<<endl;
    cout<<"6+12*2-8="<<6+12*2-8<<endl;
    cout<<"4+8*2="<<4+8*2<<endl;
    cout<<"6+17%3-2="<<6+17%3-2<<endl;
    cout<<"2+22*(9-7)="<<2+22*(9-7)<<endl;
    cout<<"(8+7)*2="<<(8+7)*2<<endl;
    cout<<"(16+7) %2-1="<<(16+7)%2-1<<endl;
    cout<<"(19-3)*(2+2)/4="<<(19-3)*(2+2)/
4<<endl;
    return 0;
}

```

```
28/4-2=5
6+12*2-8=22
4+8*2=20
6+17%3-2=6
2+22*(9-7)=46
(8+7)*2=30
(16+7) %2-1=0
(19-3)*(2+2)/4=16

[Program finished]
```

Task-03

```
#include<iostream>
using namespace std;
int main()
{
  int units;
  flaat mass;
  double weight;
cin>>units>>mass;
weight = mass *units;
cout<<"weight="<<weight<<endl;
return 0;
}
```

output:

units = 2

mass = 3.5

weight = 7

```
1  #include <iostream>
2  using namespace std;
3  int main (){
4      int units;
5      float mass;
6      double weight;
7      cin>>units>>mass;
8      weight=mass*units;
9      cout<<"weight="<<weight<<endl;
10     return 0;
11 }
12
13
```

```
2
3.5
weight=7

[Program finished]
```

Task-04

```
#include<iostream>
using namespace std;
int main()
{
    int a, b =2;
    float c= 4.2;
    a = b * c;
    cout<<"value of a="<<a<<endl;

    return 0;
}
```

Output:-

value of a = 8

```cpp
#include <iostream>
using namespace std;
int main (){
    int a,b=2;
    float c=4.2;
    a=b*c;
    cout<<"value of a="<<a<<endl;
    return 0;
}
```

value of a=8

[Program finished]

Task-05

```
#include <iostream>
 using namespace std;
  int main()
{  int qty=10;
   int salesReps=3;
   float unitsEach;
unitsEach= static_cast<float>
(qty)/salesReps;
 cout << "unitsEach="
<< unitsEach << endl;
   return 0;
}
```

Output:-

unitsEach = 3.33333

```
#include <iostream>
using namespace std;
int main (){
    int qty =10;
    int salesReps=3;
    float unitsEach;
    unitsEach=static_cast<float>
    (qty)/salesReps;
    cout<<"units Each="
    <<unitsEach<<endl;

    return 0;
}
```

units Each=3.33333

[Program finished]

Task_06

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main()
{
 srand(time(0));
 int num1 = rand() % 500 + 1;
 int num2 = rand() % 500 + 1;
 int answer;
 cout << num1 << endl;
 cout << "+" << num2 << endl;
 cout << "--------" << endl;
 cout << "press Enter to see the answer.";
  cin.get ();
   answer = num1 + num2;
   cout << num1 << endl;
   cout << "+" << num2 << endl;
   cout << answer << endl;
 return 0;
}
```

output:-

```
397
+218
------
Press Enter to see the answer
397
+218
615
```

```cpp
#include <iostream>
#include<cstdlib>
#include<ctime>
using namespace std;
int main ()
{
    srand(time(0));
    int num1=rand()%500+1;
    int num2=rand()%500+1;
    int answer;
    cout<<num1<<endl;
    cout<<"+"<<num2<<endl;
    cout<<"_ _ _ _ _"<<endl;
    cout<<"press Enter to see the answer.";
    cin.get();
    answer=num1+num2;
    cout<<num1<<endl;
    cout<<"+"<<num2<<endl;
    cout<<answer<<endl;
  return 0;
}
```

397
+218
_ _ _ _ _
press Enter to see the answer.
397
+218
615

[Program finished]

Home Task 1(A)

```
# include<iostream>
 using namespace std;
 int main()
 {
 double number1, number2, sum;
 cout << "Enter a number:";
  cin >> number1;
 Cout << "Enter another number:";
 cin>>number2;

 SUM = number1 + number2;
 cout<< "The sum of the two numbers
is "<<sum<<endl;
 return 0;
 }
```

Output:-

```
Enter a number: 5
Enter another number: 7
The sum of the two numbers is 12
```

```
1  #include <iostream>
2  using namespace std;
3  int main ()
4  {
5    double number1,number2,sum;
6    cout<<"Enter a number:";
7    cin>>number1;
8    cout<<"Enter another number:";
9    cin>>number2;
10   sum=number1+number2;
11   cout<<"The sum of the two numbers
   is"<<sum<<endl;
12   return 0;
13 }
14
15
16
```

```
Enter a number:5
Enter another number:7
The sum of the two numbers is12

[Program finished]
```

Home Task 1 (B)

```
#include<iostream>
using namespace std;
int main()
{
  int number1, number2;
  float quotient;
cout<<"Enter two numbers and I will
divide\n";
  cout<<" the first by the second for you.\n";
  cin>>number1>>number2;
  quotient = static_cast<float>(number1)/
number2;
  cout<<"Quotient = "<<quotient<<endl;
  return 0;
}
```

Output :-

```
Enter two numbers and I will divide
the first by the second for you.
7 2
Quotient = 3.5
```

```cpp
#include <iostream>
using namespace std;
int main ()
{
  int number1,number2;
  float quotient;
  cout<<"Enter two numbers and i wil divide \n";
  cout<<"the first by th second for you.\n";
cin >>number1>>number2;
  quotient=static_cast<float>(number1)/number2;
  cout<<"Quotient="<<quotient<<endl;
  return 0;
}
```

```
Enter two numbers and i wil divide
the first by th second for you.

7
2
Quotient=3.5

[Program finished]
```

Home Task 2

```
#include<iostream>
using namespace std;
int main()
 int num1 = 28;
 int num2 = 32;
 int num3 = 37;
 int num4 = 24;
 int num5 = 85;

 int sum;
 float average;
 sum = num1+num2+num3+num4+num5;
  average = sum/5.0;
  cout<<"Sum = "<<sum<<endl;
  cout<<"Average"<<average<<endl;
  return 0;
 }
```

output:-

```
Sum = 146
Average = 29.2
```

```cpp
#include <iostream>
using namespace std;
int main ()
{
int num1=18;
int num2=32;
int num3=37;
int num4=24;
int num5=35;
int sum;
float average;
sum=num1+num2+num3+num4+num5;
average=sum/5.0;
cout<<"sum="<<sum<<endl;
cout<<"Average"<<average<<endl;
  return 0;
}
```

```
sum=146
Average29.2

[Program finished]
```